

Gypsum Pulse Injection Example

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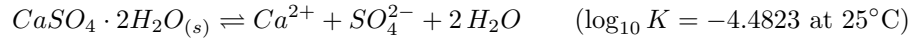
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1 Chemistry

We consider a reactive transport problem in a one-dimensional column initially containing gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) in equilibrium with resident water of low salinity. A high-concentration calcium-chloride pulse (equivalent to dissolved CaCl_2) is imposed as an initial condition in one interior cell of the column, while clean water in equilibrium with gypsum enters through the inflow boundary. This problem tests the ability of the Water Mixing Approach (WMA) to transport reactive solutes in a system governed purely by equilibrium reactions, with no kinetic reactions. The aqueous species considered are Ca^{2+} , Cl^- , SO_4^{2-} , and H_2O . The mineral is gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), which is present as a pure solid phase with constant activity equal to one. Throughout this document we assume ideal activities for the aqueous species ($\gamma_i = 1$), so that activities and molal concentrations coincide ($a_i = c_i$).

We consider the following equilibrium reaction:

- Gypsum dissolution/precipitation:



The equilibrium constant $\log_{10} K = -4.4823$ is taken from the thermodynamic database file `master25_modif.dat` used by the simulation code, which follows the standard convention of treating H_2O as constant activity ($a_{\text{H}_2\text{O}} = 1$) and absorbing it into K ; equivalently, the mass-action expression for gypsum reads $K = a_{\text{Ca}^{2+}} a_{\text{SO}_4^{2-}}$. There are no kinetic reactions in this problem ($\mathbf{S}_K = \mathbf{0}$). Since $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}_{(s)}$ and H_2O both have constant activity, the variable activity species are Ca^{2+} , Cl^- , and SO_4^{2-} .

Both in the domain (column interior) and at the inflow boundary, the water is in equilibrium with gypsum, so only the gypsum reaction applies in both cases. Consequently $\mathbf{S}_{e,v}^{\text{dom}} = \mathbf{S}_{e,v}^{\text{inf}}$, and the equilibrium stoichiometric matrix, restricted to variable activity species, is

$$\mathbf{S}_{e,v} = \begin{pmatrix} \text{Ca}^{2+} & \text{Cl}^- & \text{SO}_4^{2-} \\ 1 & 0 & 1 \end{pmatrix},$$

where the single row corresponds to gypsum dissolution/precipitation: producing one Ca^{2+} , leaving Cl^- unaffected, and producing one SO_4^{2-} .

1.1 Component matrix

Following De Simoni et al. (2005), the component matrix $\mathbf{U}$ is defined such that $\mathbf{U}\mathbf{S}_{e,v}^T = \mathbf{0}$, ensuring that the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ are conservative quantities not affected by equilibrium reactions. Here $\mathbf{S}_{e,v}$ denotes the equilibrium stoichiometric matrix restricted to variable activity species (i.e. excluding constant activity species), which in this problem are H_2O and the mineral phase $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}_{(s)}$.

Splitting $\mathbf{S}_{e,v} = (\mathbf{S}'_{e,v} \mid \mathbf{S}''_{e,v})$ into blocks corresponding to primary and secondary variable activity species, and redefining so that $\mathbf{S}''_{e,v} = -\mathbf{I}$, the component matrix takes the form

$$\mathbf{U} = (\mathbf{I}_{N_u} \mid \mathbf{S}'_{e,v}^T), \quad (1.1)$$

where $N_u = N_v - N_r$ is the number of components, N_v is the number of variable activity species and N_r the number of equilibrium reactions.

In the domain, we have $N_v = 3$ variable activity species (Ca^{2+} , Cl^- , SO_4^{2-}) and $N_r = 1$ equilibrium reaction (gypsum dissolution/precipitation), giving $N_u = 3 - 1 = 2$ components. We choose SO_4^{2-} as the secondary species (it has coefficient +1 in the gypsum dissolution/precipitation reaction), so the primary species are Ca^{2+} and Cl^- .

37 The secondary block is $\mathbf{S}_{e,v}''^{dom} = (1)$, which we redefine to $-\mathbf{I}$ by multiplying the gypsum row by -1 . This yields
 38 the transformed primary block $\mathbf{S}_{e,v}'^{dom} = (-1, 0)$ (coefficients of Ca^{2+} and Cl^- after the sign change). The component
 39 matrix is therefore $\mathbf{U}^{dom} \in \mathbb{R}^{2 \times 3}$:

$$\mathbf{U}^{dom} = \left(\mathbf{I} \mid \mathbf{U}_2^{dom} \right)$$

40 where, with rows corresponding to the primary species Ca^{2+} and Cl^- ,

$$\mathbf{U}_2^{dom} = (\mathbf{S}_{e,v}'^{dom})^T = \begin{pmatrix} SO_4^{2-} \\ -1 \\ 0 \end{pmatrix}.$$

41 Written out explicitly in the natural species order,

$$\mathbf{U}^{dom} = \begin{pmatrix} Ca^{2+} & Cl^- & SO_4^{2-} \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},$$

42 which yields the conservative components

$$\begin{aligned} u_1^{dom} &= c_{Ca^{2+}} - c_{SO_4^{2-}}, \\ u_2^{dom} &= c_{Cl^-}. \end{aligned}$$

43 The physical interpretation is straightforward: gypsum dissolution/precipitation produces equal moles of Ca^{2+} and
 44 SO_4^{2-} , so their difference u_1 is invariant under this reaction; and the reaction does not affect Cl^- , so u_2 is also
 45 unaffected by it. Both components are still subject to advection and dispersion — they are conservative *with respect to*
 46 *the chemical reactions*, not constant in space and time.

47 **Component matrix of the boundary (inflow) water.** At the inflow, the same gypsum equilibrium applies and
 48 no additional reactions are active, so the set of variable activity species, the equilibrium stoichiometric matrix and the
 49 choice of primary/secondary species are all identical to those of the domain. Consequently the chemistry of the inflow
 50 water is identical to that of the domain water, and the boundary component matrix coincides with the domain one:

$$\mathbf{U}^{inf} = \mathbf{U}^{dom} = \begin{pmatrix} Ca^{2+} & Cl^- & SO_4^{2-} \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

51 The two components at the inflow are therefore $u_1^{inf} = c_{Ca^{2+}} - c_{SO_4^{2-}}$ and $u_2^{inf} = c_{Cl^-}$, with the same expressions as
 52 in the domain. This contrasts with problems where additional inflow-only equilibria are active (e.g. atmospheric gas
 53 equilibria), in which case $\mathbf{U}^{inf}$ would have fewer rows than $\mathbf{U}^{dom}$.

54 **Note on the component matrix used in the simulation.** The reactive transport simulation uses the domain
 55 component matrix $\mathbf{U}^{dom}$. The component concentrations $\mathbf{u} = \mathbf{U}^{dom} \mathbf{c}$ used in the reactive transport simulation are
 56 always computed with the domain component matrix, regardless of whether the underlying species concentrations $\mathbf{c}$
 57 correspond to the domain (initial or interior) or to the boundary (inflow) water. In this particular problem $\mathbf{U}^{inf} = \mathbf{U}^{dom}$,
 58 so the projection $\mathbf{u}_{inf} = \mathbf{U}^{dom} \mathbf{c}_{inf}$ coincides with $\mathbf{U}^{inf} \mathbf{c}_{inf}$; nevertheless, the rule is applied consistently with related
 59 example problems (e.g. the denitrification and calcite-equilibrium documents) where the two matrices differ.

60 The initial and boundary concentrations we consider are in Table 1. The initial water in the column is a chloride-free
 61 water in equilibrium with gypsum (the chloride concentration is set to a numerical zero of 10^{-16} mol/kg_w), except in
 62 one interior cell (cell 11 out of 30) where a high-concentration $CaCl_2$ pulse is imposed (equivalent to dissolved $CaCl_2$
 63 saturated with gypsum). The inflow water has the same chemistry as the resident water. All three waters satisfy the
 64 gypsum equilibrium constraint; only the pulse water is, in addition, charge-balanced.

	initial (resident)	pulse (cell 11)	inflow (boundary)
Ca^{2+}	1.0000329371	200.000000165	1.0000329371
Cl^-	10^{-16}	400	10^{-16}
$SO_4^{2- \dagger}$	$3.29371 \cdot 10^{-5}$	$1.64691 \cdot 10^{-7}$	$3.29371 \cdot 10^{-5}$

Table 1: Primary species concentrations of initial, pulse, and boundary waters (in mol/kg_w). All three waters are saturated with gypsum; the pulse water is a charge-balanced $CaCl_2$ solution, while the resident and inflow waters contain only a numerical-zero chloride concentration. [†]Under the ideal-activity assumption ($\gamma_i = 1$, $a_i = c_i$), the SO_4^{2-} concentrations are imposed by equilibrium with gypsum, $c_{Ca^{2+}} c_{SO_4^{2-}} = 10^{-4.4823} \approx 3.2937 \cdot 10^{-5}$. In the pulse water, $c_{Ca^{2+}}$ is set so that the water is electroneutral, $2c_{Ca^{2+}} = c_{Cl^-} + 2c_{SO_4^{2-}}$ (equivalently, $u_1 = c_{Ca^{2+}} - c_{SO_4^{2-}} = 200$ exactly); in the resident and inflow waters, $c_{Ca^{2+}}$ is set so that $u_1 = c_{Ca^{2+}} - c_{SO_4^{2-}} = 1$ exactly (these waters are not electroneutral, since their chloride is a numerical zero). Note that in the high- Ca^{2+} pulse cell, the gypsum equilibrium drives SO_4^{2-} to a much lower value (common-ion effect), while in the low- Ca^{2+} resident and inflow water SO_4^{2-} is higher.

In terms of components (same units as the species concentrations, mol/kg_w), this translates into

$$\mathbf{u}_{init} = \begin{pmatrix} 1.00000 \\ 10^{-16} \end{pmatrix}, \quad \mathbf{u}_{pulse} = \begin{pmatrix} 200.00000 \\ 400.00000 \end{pmatrix}, \quad \mathbf{u}_{inf} = \begin{pmatrix} 1.00000 \\ 10^{-16} \end{pmatrix},$$

i.e. the components in the resident and inflow water are identical (both are essentially chloride-free waters in equilibrium with gypsum, with $u_1 = 1$ exactly), while the pulse cell has a first component two hundred times larger and carries virtually all of the chloride mass in the system.

1.2 Mineral mass balance

In addition to the aqueous mass balance (2.1), the gypsum mineral content m_{gyp} , expressed in moles of gypsum per unit volume of porous medium, evolves only through dissolution/precipitation (no transport, since the mineral is immobile):

$$\frac{\partial m_{gyp}}{\partial t} = -\phi r_e, \quad (1.2)$$

where r_e is the rate of the gypsum dissolution/precipitation reaction as written in Section 1, expressed in moles per unit time per unit volume of water (positive when gypsum dissolves, producing aqueous Ca^{2+} and SO_4^{2-}); the porosity factor ϕ converts it to moles per unit time per unit volume of porous medium, consistently with the definition of m_{gyp} . The initial mineral content corresponds to a volume fraction of 0.50 throughout the column, ensuring that gypsum is present everywhere and the equilibrium constraint can be enforced locally at all times during the simulation.

2 Transport

We solve the following set of equations:

$$\begin{cases} \phi \frac{\partial \mathbf{c}^T}{\partial t} = \mathcal{L}^T(\mathbf{c}) + \phi \mathbf{r}^T \\ \mathbf{S}_e \log \mathbf{c} = \log \mathbf{K}_e \\ \mathbf{r}_K = \mathbf{r}_K(\mathbf{c}) \end{cases} \quad (2.1)$$

where $\phi \in (0, 1]$ is porosity; $\mathbf{c} \in \mathbb{R}^{n_s}$ is the concentration array of all species (n_s is the number of species); $\mathcal{L}^T(\mathbf{c}) = -\mathbf{q}^T \nabla \mathbf{c}^T + \nabla \cdot (\mathbf{D} \nabla \mathbf{c}^T) + f_w (\mathbf{c}_r^T - \mathbf{c}^T)$ is the transposed advective form of the transport operator, if we solve transport using the advective ADE, where $\mathbf{q} \in \mathbb{R}^d$ is the Darcy flux, expressed as volume of water per unit area of porous medium per unit time (d is the Euclidean dimension, in this case $d = 1$), $\mathbf{D} \in \mathbb{R}^{d \times d}$ is the hydrodynamic dispersion tensor, defined on a Darcy basis (i.e. built from $\mathbf{q}$ rather than the pore velocity, which is why no porosity factor multiplies the dispersion term), f_w is the (volumetric) water sink-source, expressed as volume of water per unit volume of porous medium per unit time, with concentration $\mathbf{c}_r$ if $f_w > 0$ (recharge), or, $\mathbf{c}_r = \mathbf{c}$ if $f_w < 0$ (discharge), or $\mathbf{c}_r = 0$ when $f_w < 0$ represents evaporation; $\mathbf{r} = \mathbf{S}_e^T \mathbf{r}_e + \mathbf{S}_K^T \mathbf{r}_K$ is the contribution of chemical reactions to the mass balance of species, expressed in moles per unit time per unit volume of water (hence the porosity factor multiplying $\mathbf{r}^T$ in (2.1)); $\mathbf{S}_e \in \mathbb{R}^{n_e \times n_s}$ and $\mathbf{S}_K \in \mathbb{R}^{n_K \times n_s}$ are the stoichiometric matrices of equilibrium and kinetic reactions, respectively, whose respective reaction rates are given by $\mathbf{r}_e \in \mathbb{R}^{n_e}$ and $\mathbf{r}_K \in \mathbb{R}^{n_K}$ (n_e and n_K are the number of equilibrium and kinetic reactions, respectively) and $\mathbf{K}_e \in \mathbb{R}^{n_e}$ are the equilibrium constants. In this problem, $n_e = 1$ (gypsum equilibrium) and $n_K = 0$, so $\mathbf{r} = \mathbf{S}_e^T \mathbf{r}_e$. All concentrations in (2.1) are expressed per unit volume of water; under the dilute-solution, unit-water-density assumption adopted here, they coincide numerically with the molal values (mol/kg_w) of Section 1.

Equation (2.1) needs to be solved subject to appropriate initial ($\mathbf{c}(\mathbf{x}, t) = \mathbf{c}(\mathbf{x}, 0)$ throughout the model domain, with a high-concentration $CaCl_2$ pulse imposed in one interior cell) and boundary conditions, which we choose to be prescribed mass flux, that is,

$$\begin{cases} \mathbf{n}^T (\mathbf{q}\mathbf{u} - \mathbf{D}\nabla\mathbf{u}) = q_{inf}\mathbf{u}_{inf} & \text{at the inflow} \\ \mathbf{n}^T (-\mathbf{D}\nabla\mathbf{u}) = 0 & \text{at the outflow} \end{cases}$$

expressing that the flux of components at the inflow is $q_{inf}\mathbf{u}_{inf}$, and that there is no dispersive flux at the outflow. In the component-based formulation actually solved by the simulation, transport is integrated on the conservative components $\mathbf{u} = \mathbf{U}^{dom}\mathbf{c}$ rather than directly on the species concentrations $\mathbf{c}$, which removes the equilibrium reaction term from the transport step.

To solve numerically, we choose a uniform mesh made of 30 cells of length $\Delta x = 1$ m (domain length $L = 30$ m), with targets at the cell centres. The pulse is imposed at cell 11. The time step is $\Delta t = 1$ d, and the simulation is run for a total time of $T = 40$ d (i.e. 40 time steps), long enough for the pulse to be advected and dispersed across the column at the chosen Darcy velocity ($qT/\phi = 40$ m $>$ L). Transport is integrated with the fully implicit Euler scheme (time weighting factor $\theta = 1$); the time weighting factor for reactions is set to 0.5 (Crank–Nicolson), although it is immaterial in this problem, since there are no kinetic reactions. We consider uniform transport properties: $\phi = 0.5$, $q = q_{inf} = 0.5$ m/d, $\alpha_L = 1$ m. The dispersion coefficient is $D = \alpha_L q = 0.5$ m²/d (molecular diffusion is neglected). Note that while the chemistry described in Section 1 (species, gypsum reaction, and equilibrium constant) is intrinsic to this gypsum equilibrium problem, the transport parameters and numerical settings — mesh size, time step, porosity, Darcy flux, dispersivity, boundary condition type, and integration scheme — can be freely modified by the user to suit different physical scenarios, subject to the following stability and accuracy constraints:

- **Grid Péclet number:** $Pe = \frac{q\Delta x}{D} = \frac{\Delta x}{\alpha_L} \leq 2$ to avoid spurious oscillations in centred schemes. In this example, $Pe = 1$.
- **Courant number:** $Co = \frac{q\Delta t}{\phi\Delta x} \leq 1$ for time integration. In this example, $Co = 0.5 \cdot 1 / (0.5 \cdot 1) = 1$, exactly at the limit.
- **Von Neumann stability:** For the explicit Euler scheme, the time step must satisfy $\Delta t \leq \frac{\phi(\Delta x)^2}{2D}$ for dispersion stability (0.5 d in this example, which the chosen $\Delta t = 1$ d would violate). The fully implicit Euler scheme is unconditionally stable, so this constraint does not apply.
- **Combined constraint (explicit Euler):** When using explicit time integration, both the Courant and dispersion constraints must be satisfied simultaneously: $\Delta t \leq \min\left(\frac{\phi\Delta x}{q}, \frac{\phi(\Delta x)^2}{2D}\right)$, which in this example gives $\Delta t \leq \min(1, 0.5) = 0.5$ d.

The fully implicit Euler scheme used in this example is unconditionally stable, allowing larger time steps at the cost of solving a nonlinear system at each step. However, accuracy still requires Δt small enough to resolve the dissolution/precipitation dynamics induced by the pulse.

References

- De Simoni, M., Carrera, J., Sánchez-Vila, X., and Guadagnini, A. (2005). A procedure for the solution of multicomponent reactive transport problems. *Water Resources Research*, 41(11):W11410. doi: [10.1029/2005WR004056](https://doi.org/10.1029/2005WR004056).